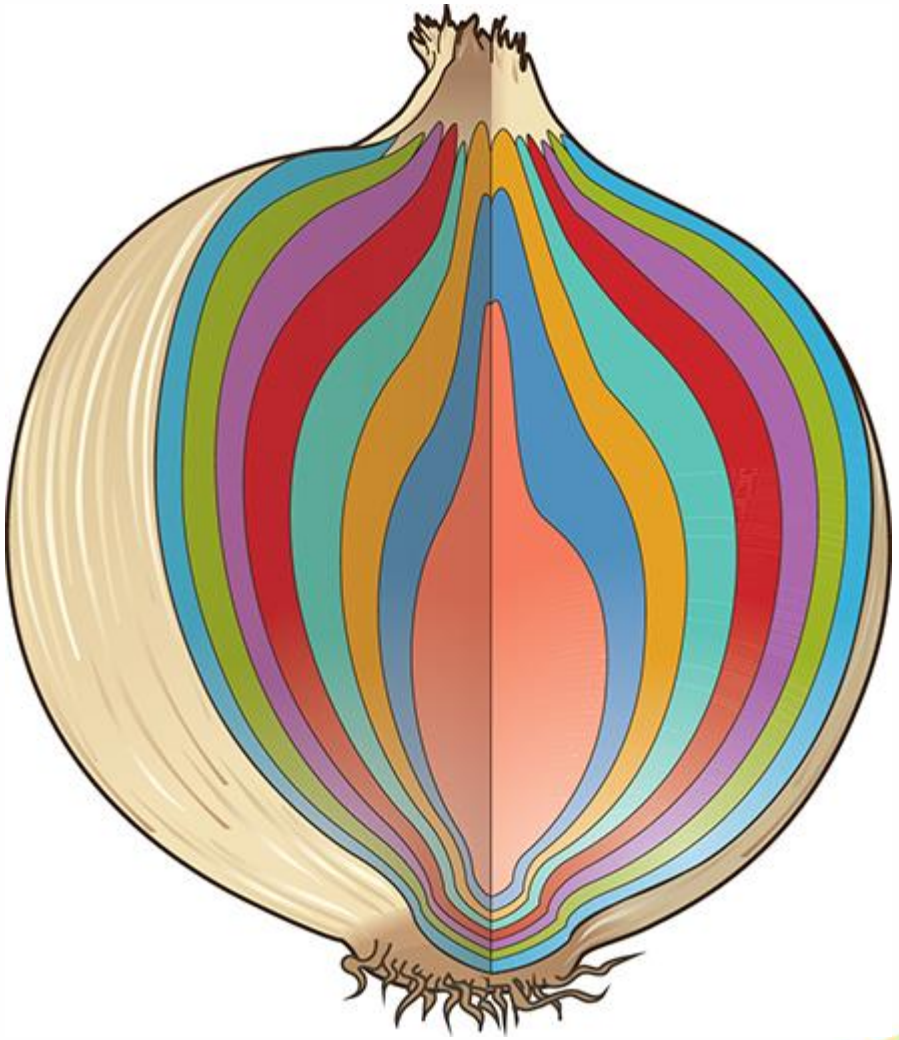
UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Basic Concepts of Information Technology

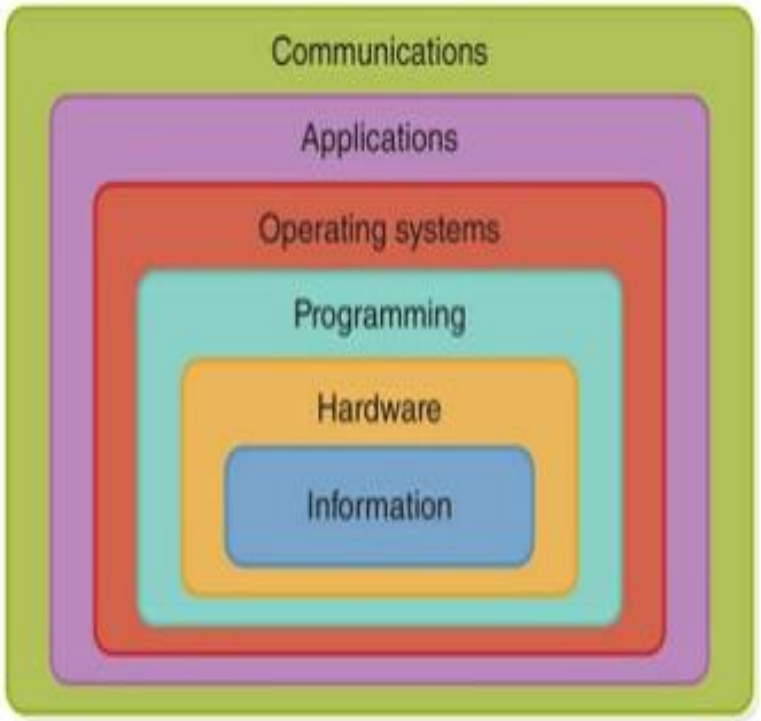
Instructor: MSc. Bui Trong Hieu

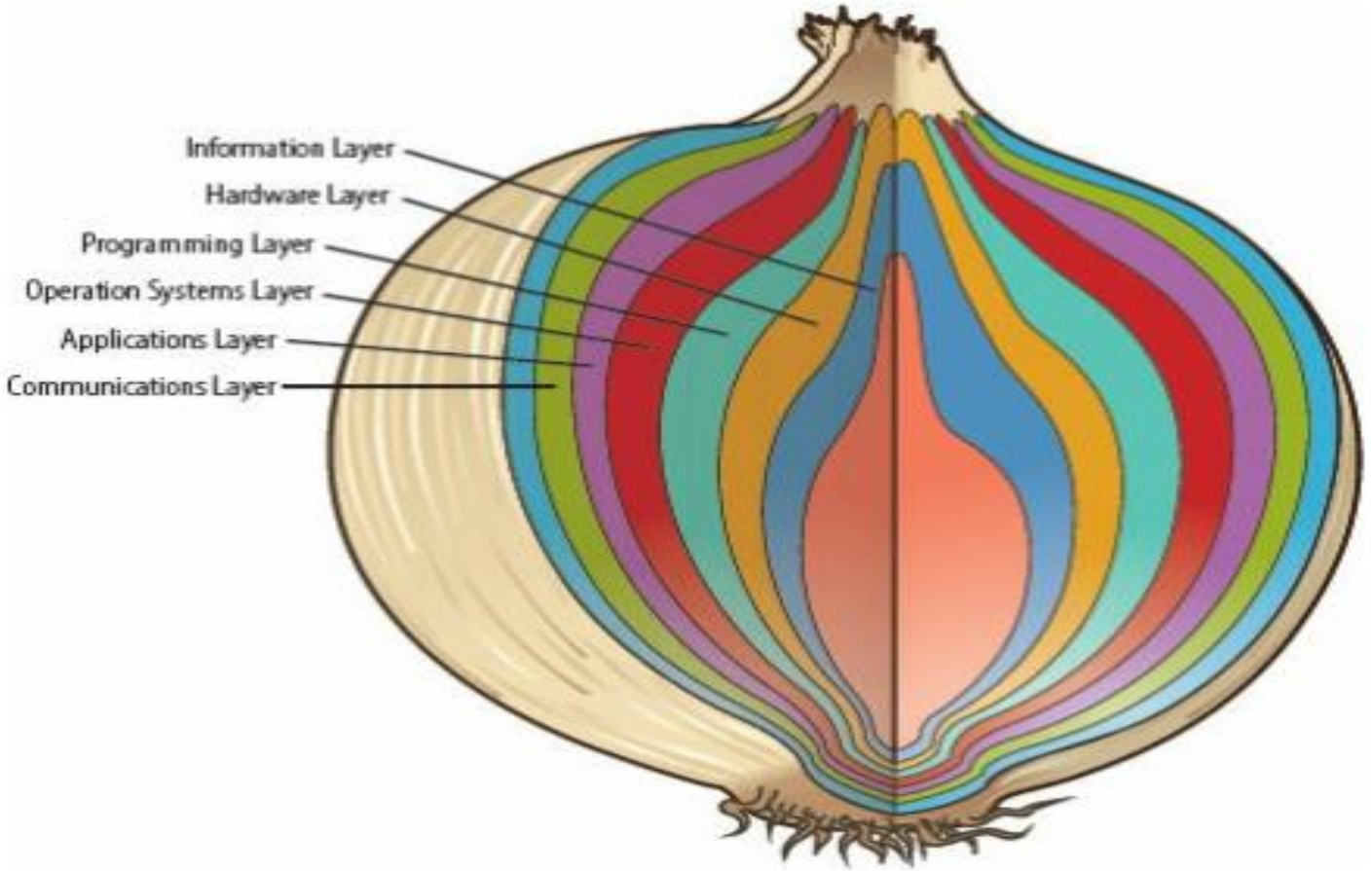
Outline

- Data Representation
- Data Processing
- Computer Systems
- Hardwares
- Softwares



Layers of a Computing System

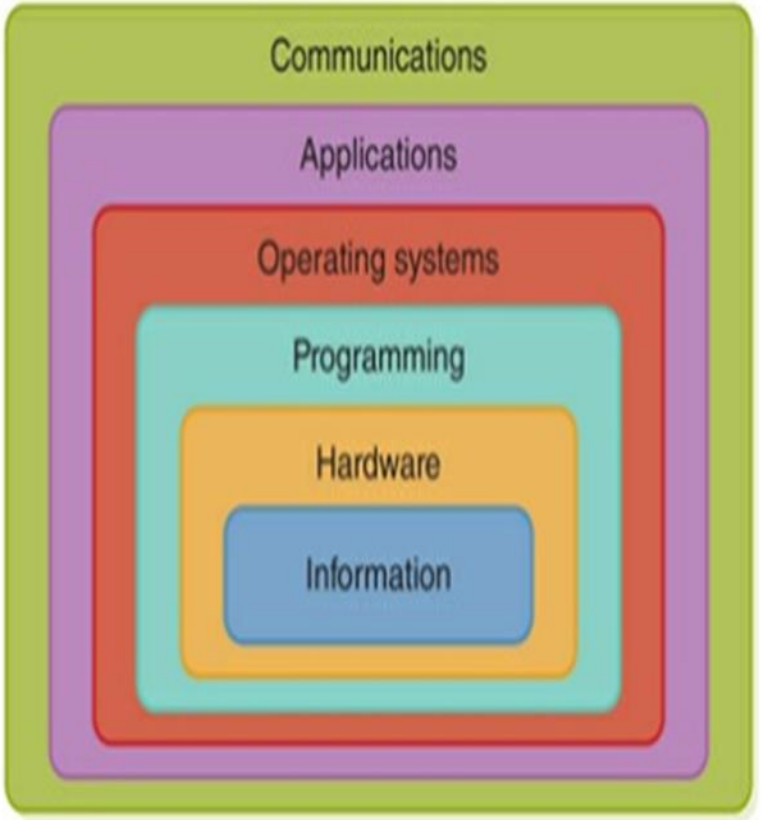




UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Inner Layers of a Computer System

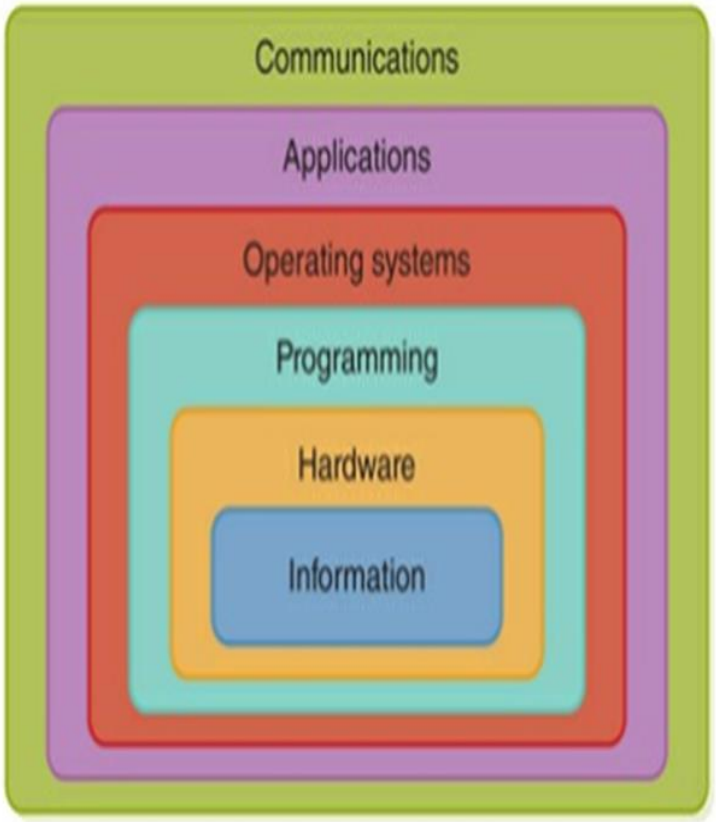
Layers	Definitions and Effects
Information layer	The way to represent information on a computer (Binary digit 1,0)
Hardware layer	Devices (gates, circuits) to control the flow of electricity→components (CPU, memory)
Program-ming layer	software, instructions, programs to solve problems
OS layer	to manage the computer's resources (to interact with many aspects of a computer)



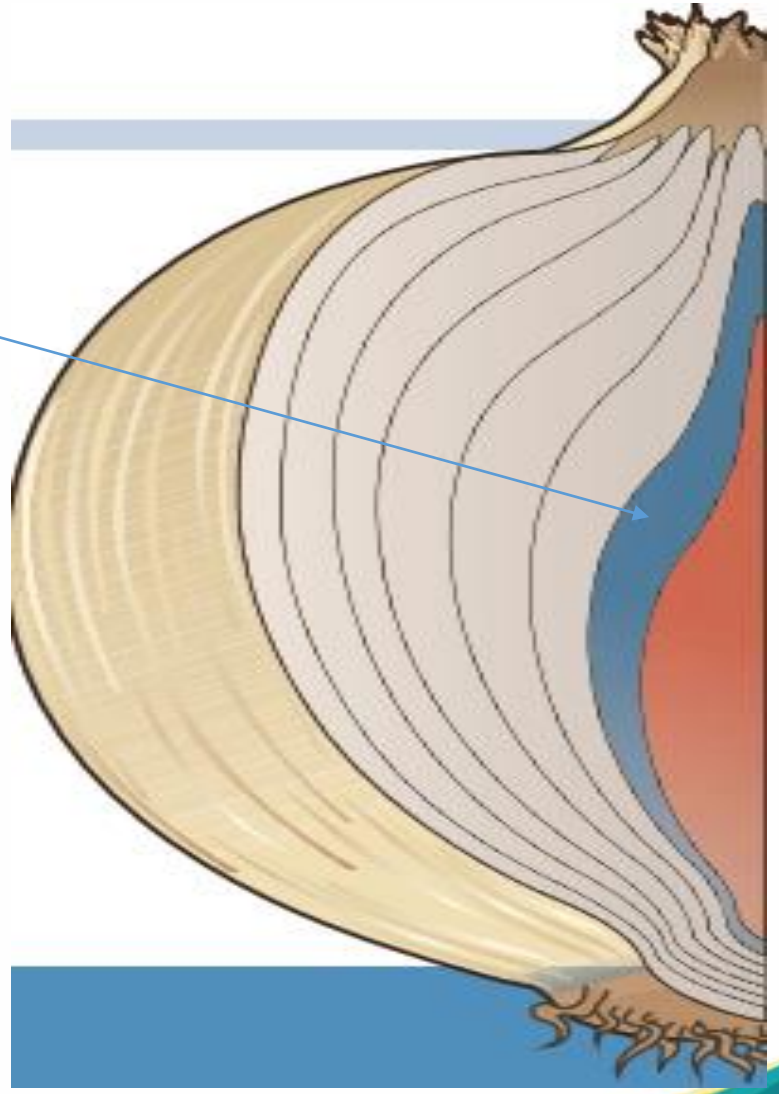
UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Outer Layers of a Computer System

Application layer	to solve specific real-world problems,
Communication layer	a fundamental layer at which computing systems operate. using computer connected into networks to communicate. ➢ Internet to share information and resources

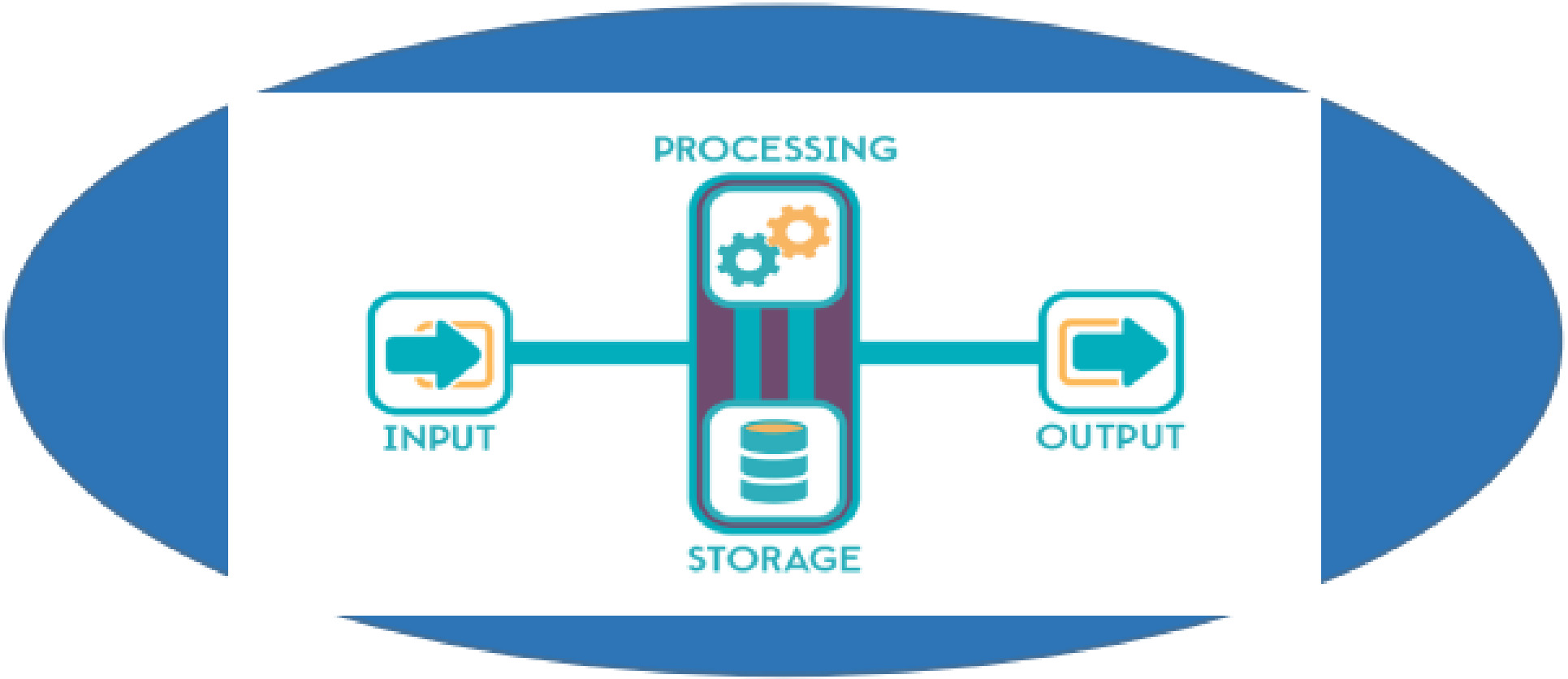


The Information Layer

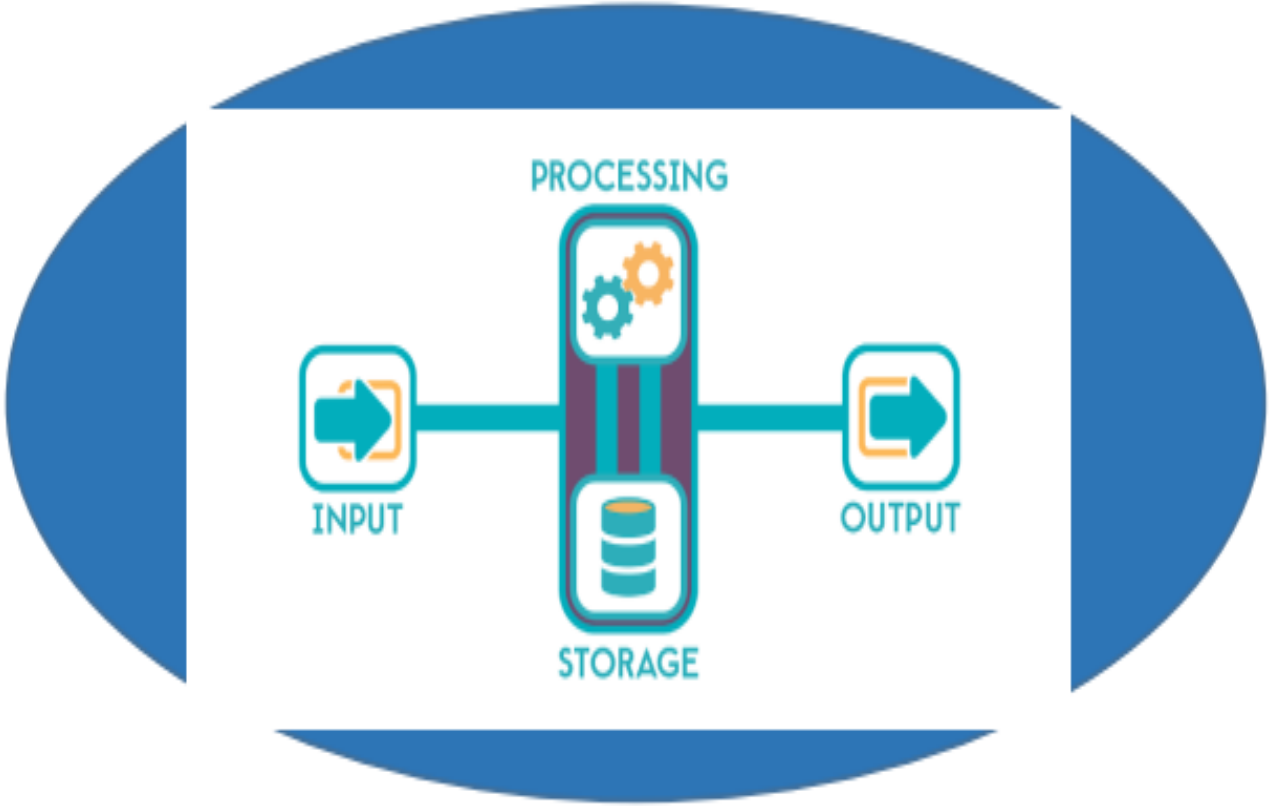


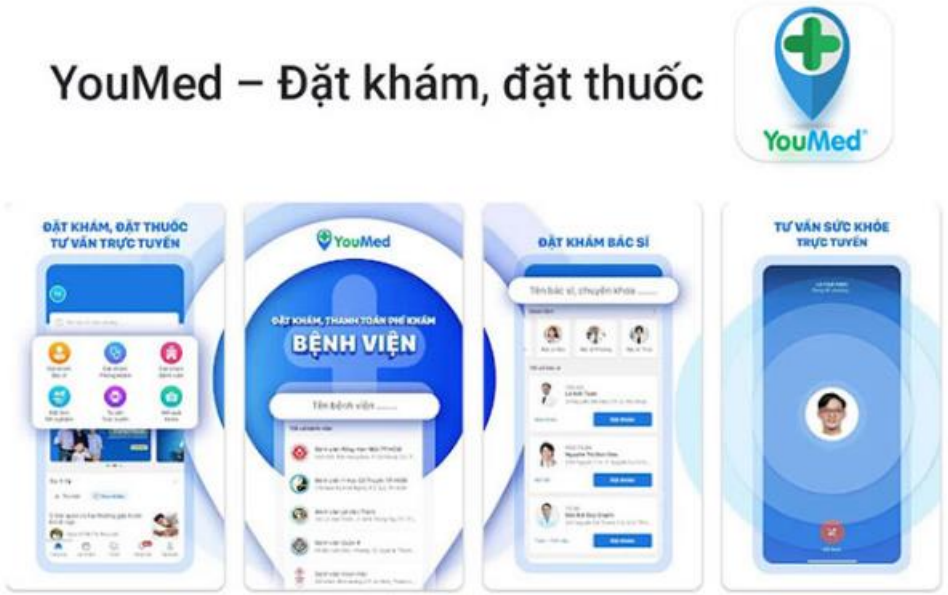
- Binary Values and Number Systems
- Data Representation
  - Numbers
  - Text, Audio, Images, Video

How Do Computers Work?

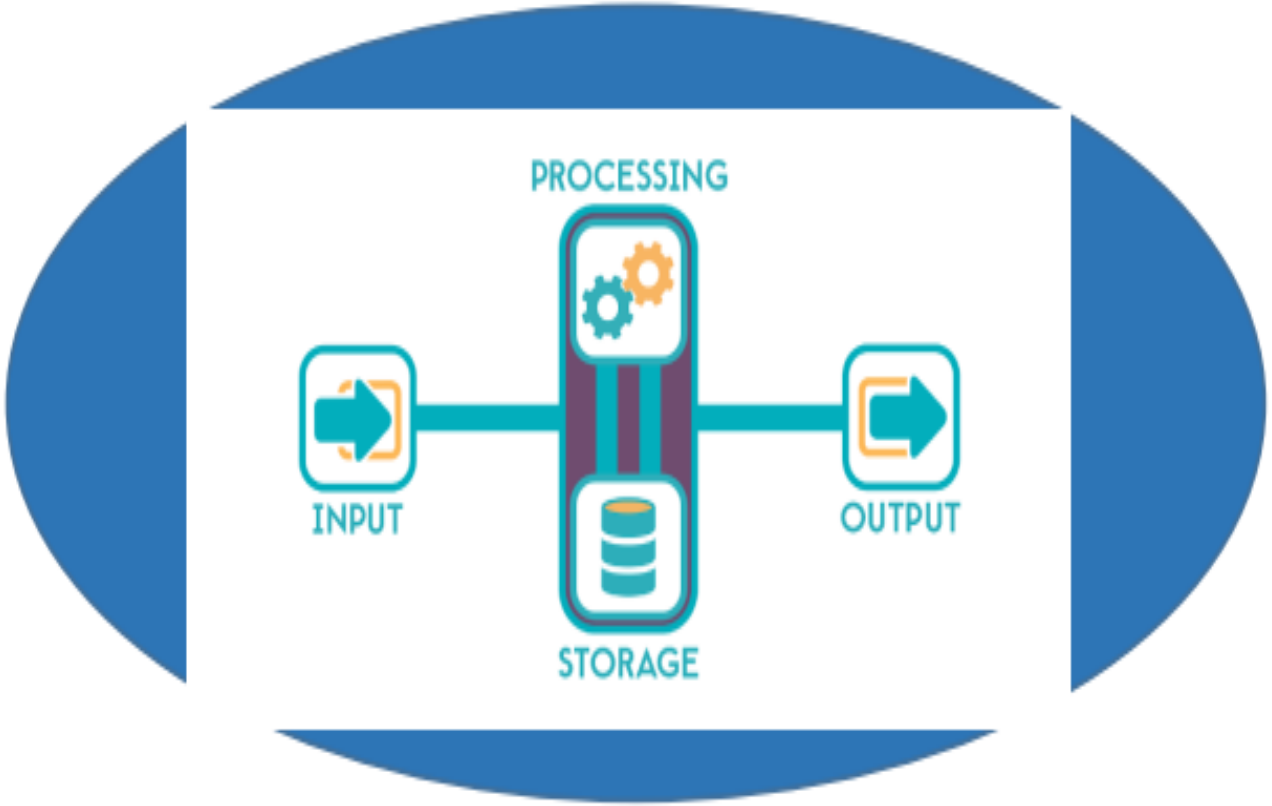


How Do Computers Work?



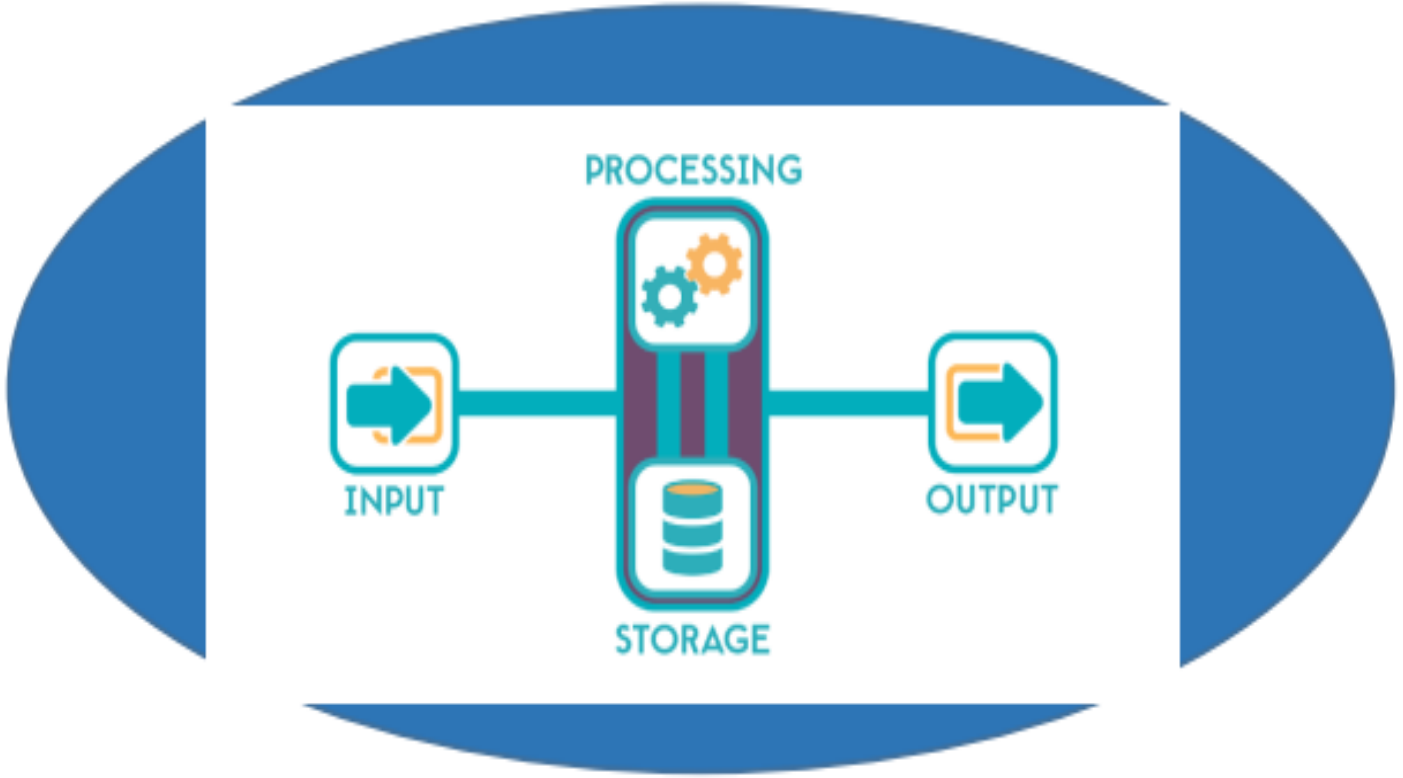


How Do Computers Work?

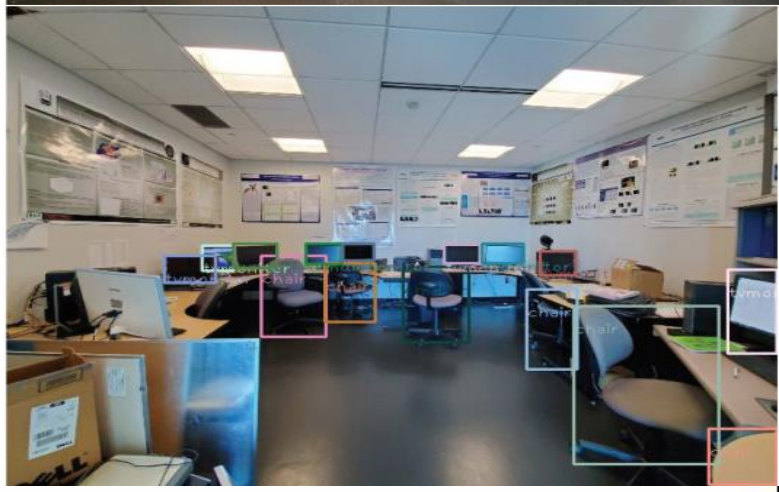




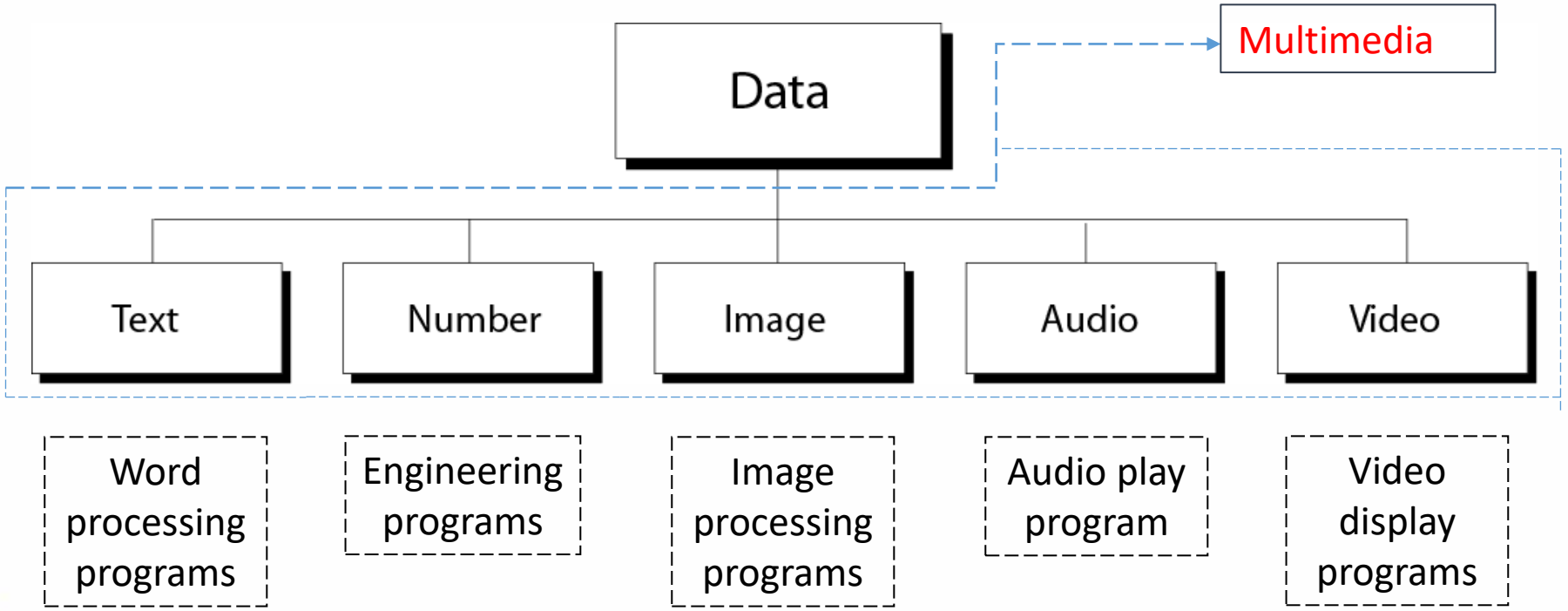
How Do Computers Work?



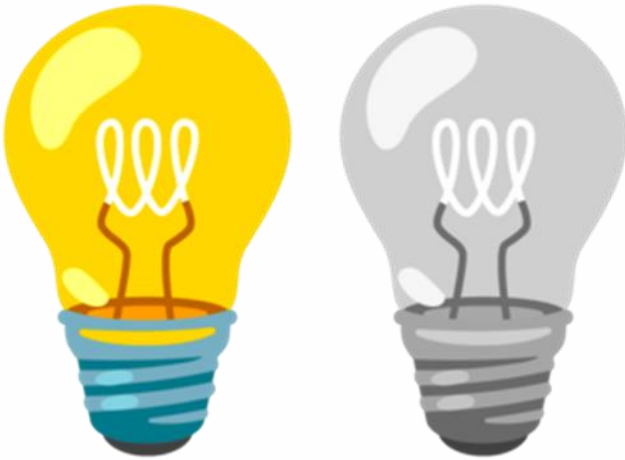




Data Types

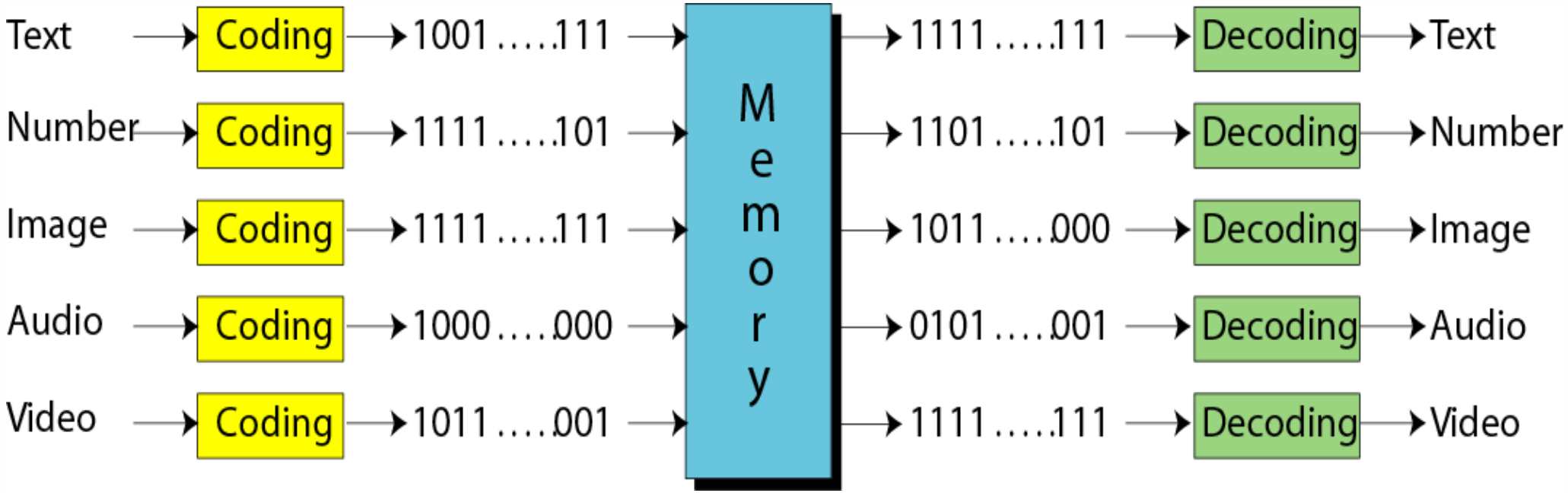

Data
Multimedia
Text
Number
Image
Audio
Video
Word processing programs
Engineering programs
Image processing programs
Audio play program
Video display programs

Representation



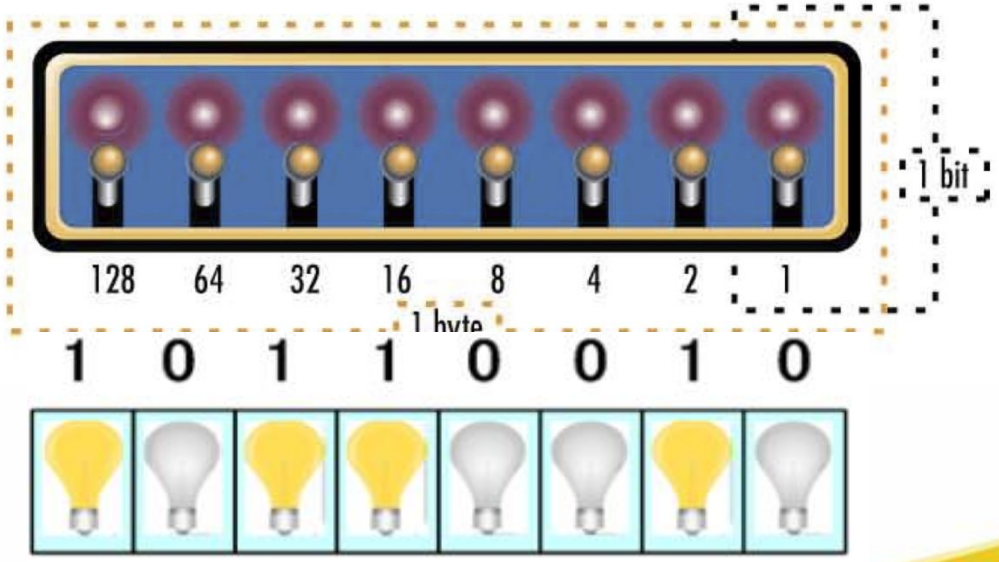
- ALL computers are electronic, digital devices
  - **Digital devices** work with discrete data, such as digits 1 and 0, or like a light switch – on or off
- These 1s and 0s are referred to as *bi*nary dig*its* or shortened to ***bits***
- Computers use sequences of bits to digitally represent numbers, letters, punctuation marks, music, pictures, and videos

Coding and Decoding



Bit Basics

- ***Bit:*** From Binary digit
  - Smallest unit of information computer can process
  - Can have one of two values: 0 or 1
- ***Byte***
  - Collection of 8 bits
  - Can represent 256 different messages ($256 = 2^8$)



Bits as Numbers

- Denotes all numbers with combinations of 0s and 1s
- Decimal numbers automatically converted to binary
- Binary number processing hidden from user

Decimal	Binary	Decimal	Binary
0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001

Bits as Codes

- Codes represent each letter, digit, and special character
- ***ASCII:*** Most widely used
  - Each character is a unique 8-bit code
  - 256 unique codes for 26 letters, 10 digits, special characters
- ***Unicode:*** Supports more than 100,000 unique characters

Character	ASCII binary code
A	01000001
B	01000010
C	01000011
D	01000100
E	01000101
F	01000110
G	01000111
H	01001000
I	01001001
J	01001010
K	01001011
L	01001100
M	01001101
N	01001110
O	01001111
P	01010000
Q	01010001
R	01010010
S	01010011
T	01010100
U	01010101
V	01010110
W	01010111
X	01011000
Y	01011001
Z	01011010
0	00110000
1	00110001
2	00110010
3	00110011
4	00110100
5	00110101
6	00110110
7	00110111
8	00111000
9	00111001

Text

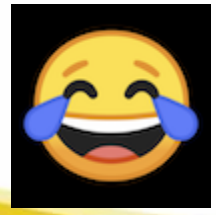
ASCII:

H	I	!
01001000	01001001	00100001
72	73	33

Unicode:

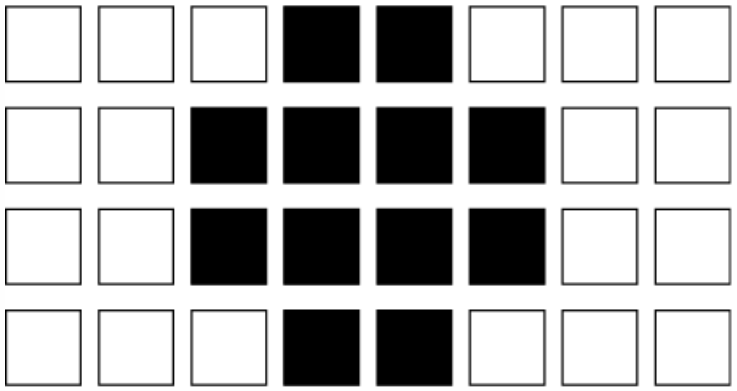
`11110000 10011111 10011000 10000010`

`4036991106`



Image

black-and-white image



Image

0 0 0 1 1 0 0 0
0 0 1 1 1 1 0 0
0 0 1 1 1 1 0 0
0 0 0 1 1 0 0 0

Matrix Representation

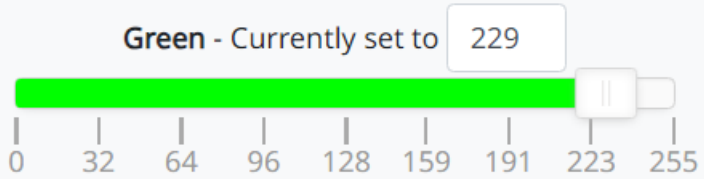
00011000 00111100 00111100 00011000

Linear Representation

Images and Colours

rgb(136,229,134)



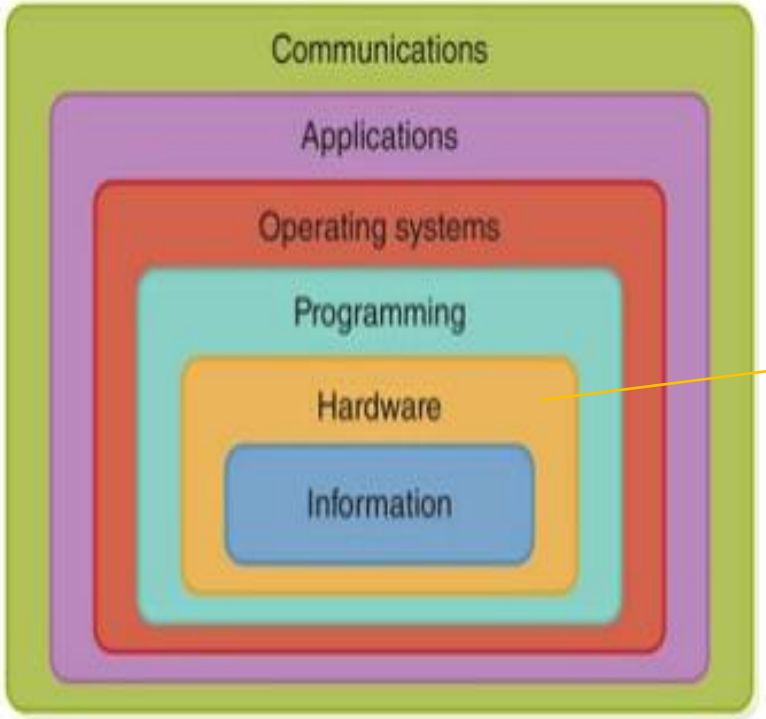


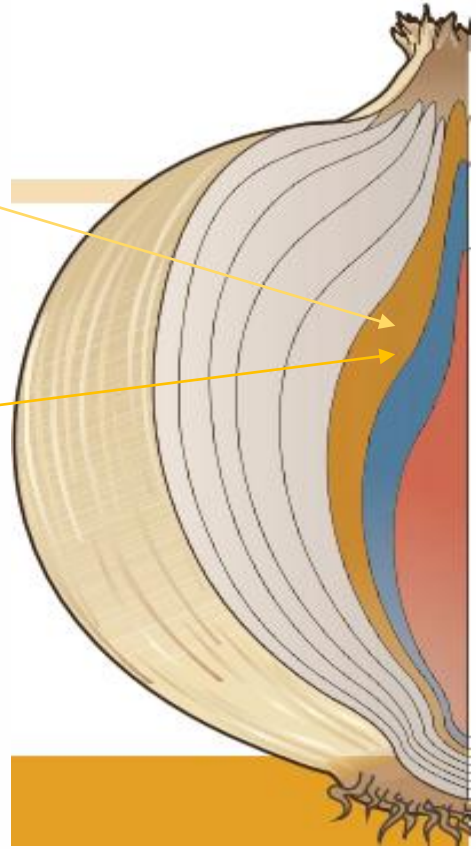


RGB

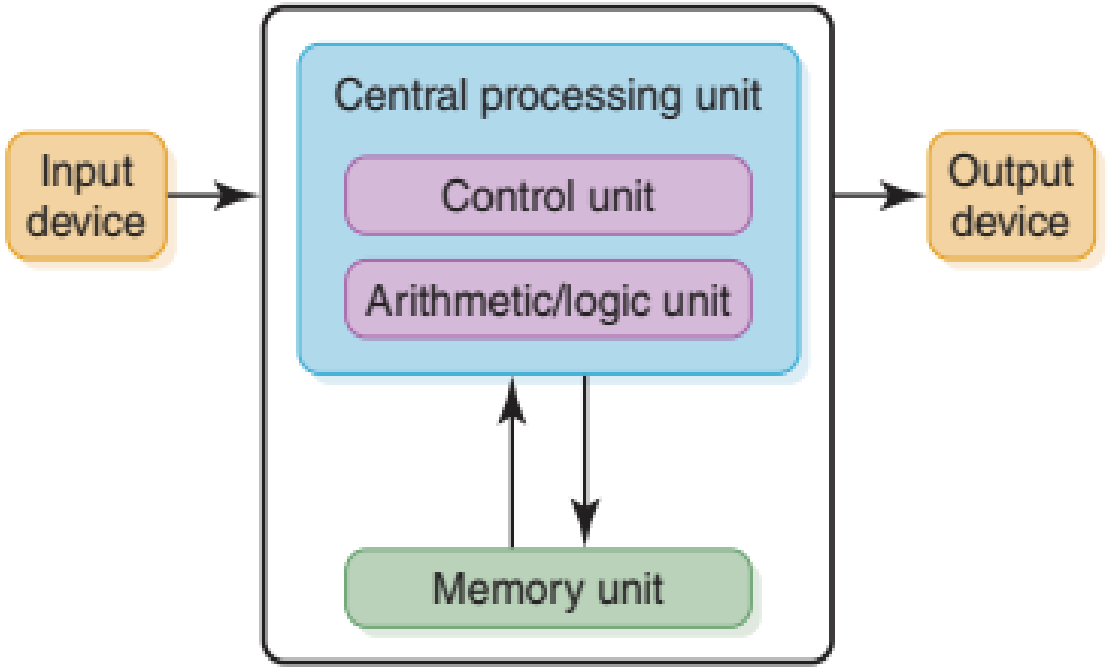
72 73 33

The Hardware Layer



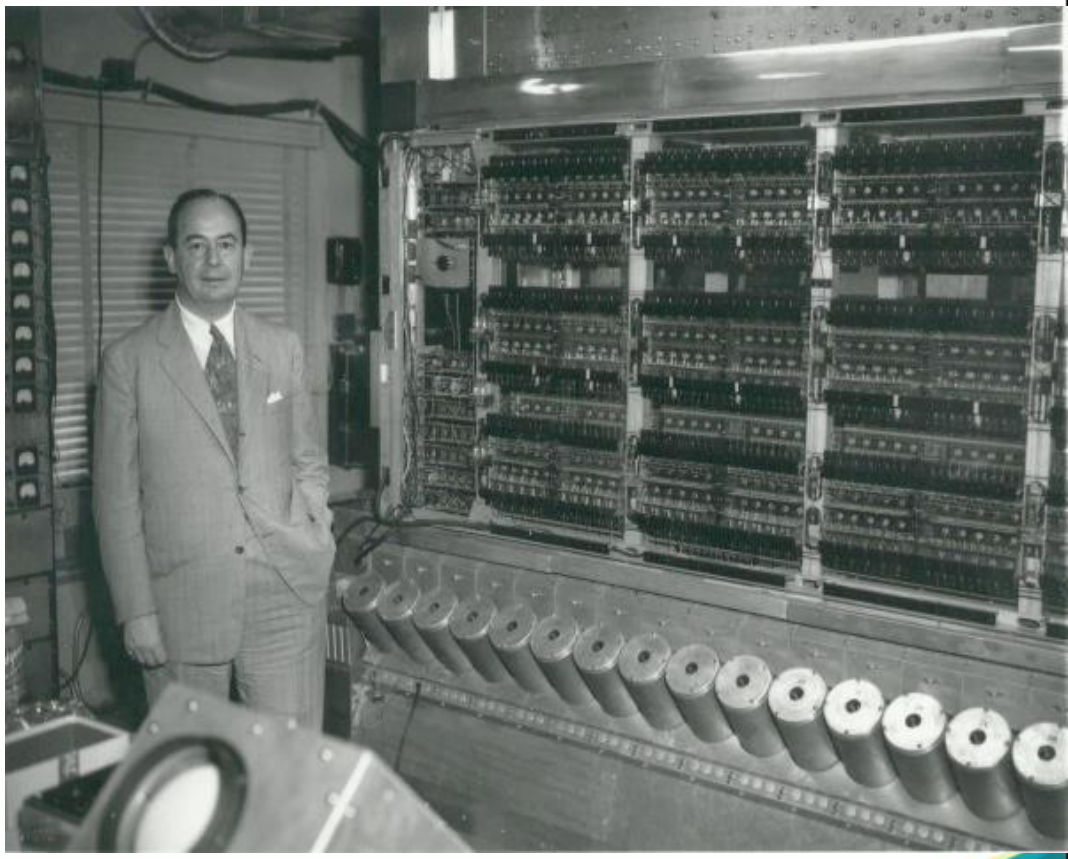


The von Neumann Architecture



The von Neumann architecture

Stored-Program Concept



John von Neumann and his machine

Memory

Address	Contents
00000000	11100011
00000001	10101001
:	:
.	.
11111100	00000000
11111101	11111111
11111110	10101010
11111111	00110011







A collection of cells, each with a unique physical address; both addresses and contents are in binary

7	6	5	4	3	2	1	0	← Bit position
1	0	1	0	1	0	1	0	← Contents

Random Access Memory (RAM)
Memory in which each location can be accessed and changed
Read Only Memory (ROM)
Memory in which each location can be accessed but *not* changed
RAM is volatile, ROM is not

What does volatile mean?

UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Secondary Storage Devices

Why is it necessary to have secondary storage devices?

Can you name some of these devices?



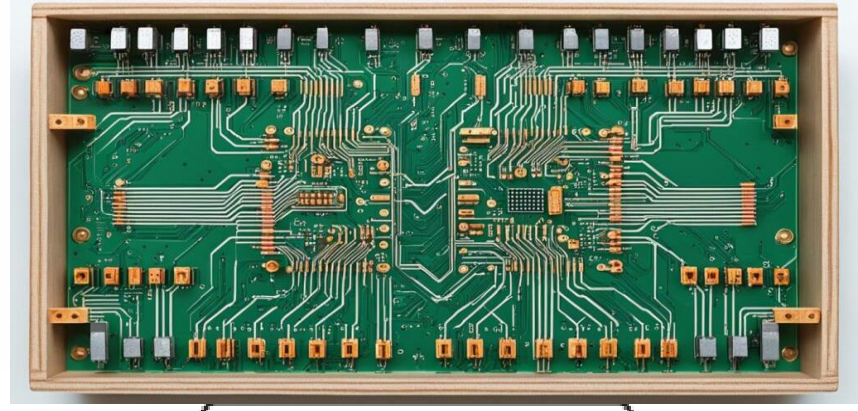
UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

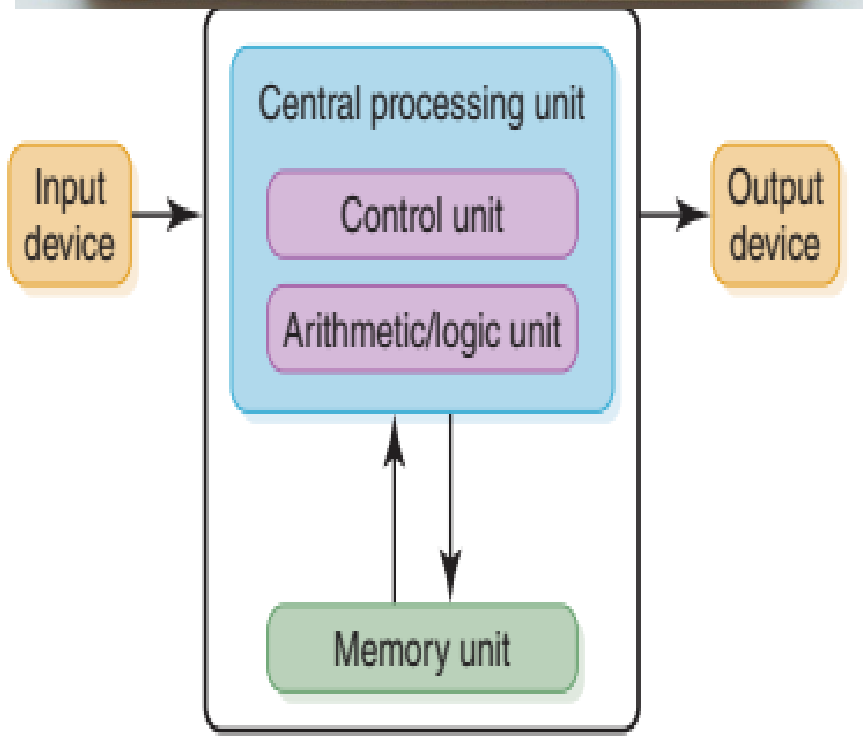
Arithmetic/Logic Unit

Performs basic arithmetic operations such as adding

Performs logical operations such as AND, OR, and NOT

Most modern ALUs have a small amount of special storage units called **registers**





The von Neumann architecture

UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Input/Output Units



Input Unit

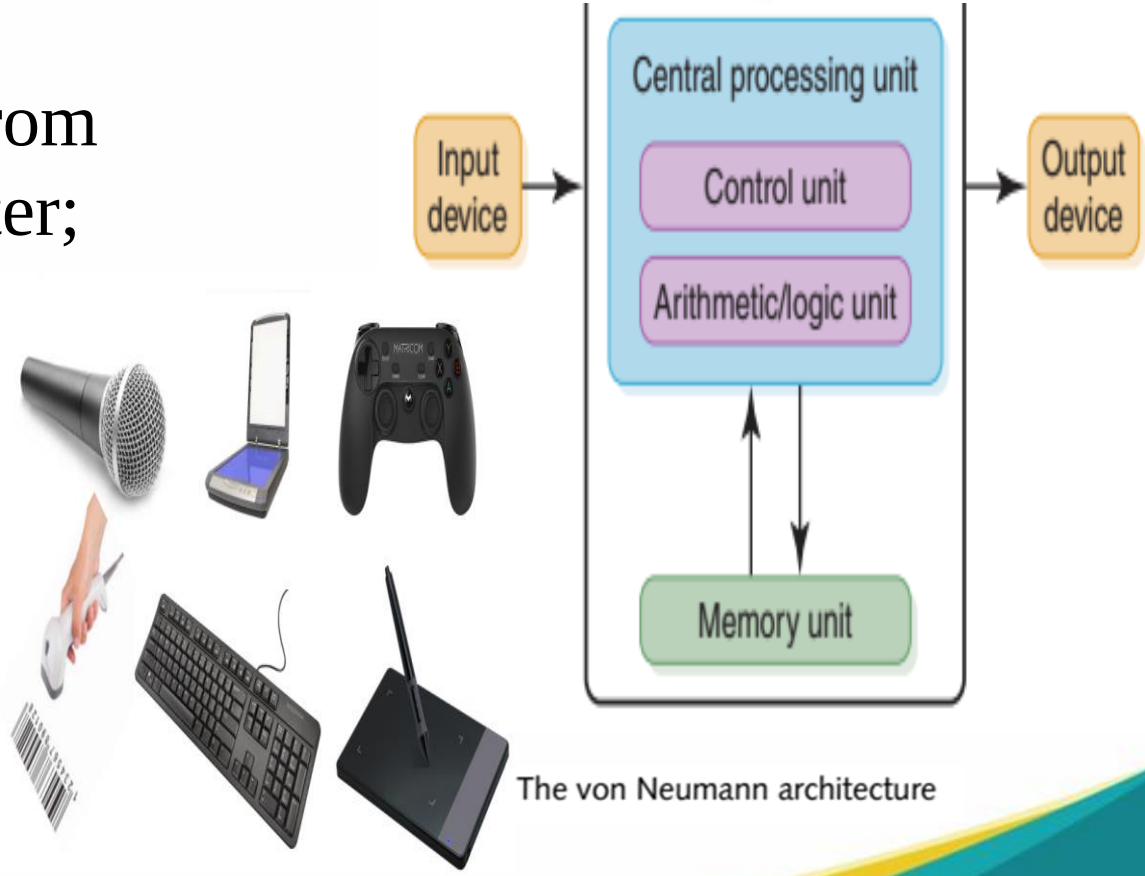
A device through which data and programs from the outside world are entered into the computer;

Can you name three?

Output unit

A device through which results stored in the computer memory are made available to the outside world

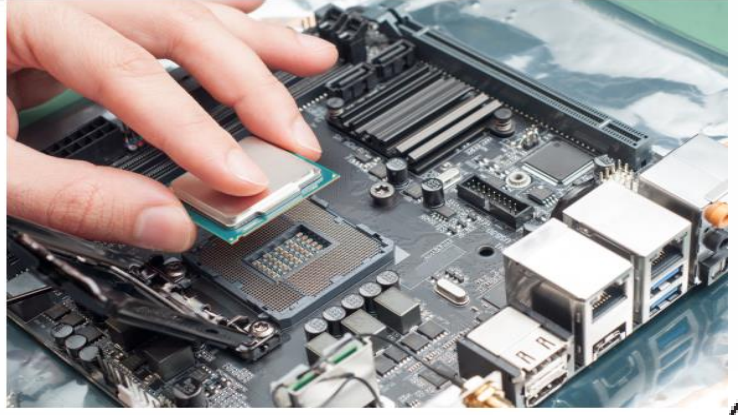
Can you name two?



The von Neumann architecture

UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Control Unit





Control unit

The organizing force in the computer

Instruction register (IR)

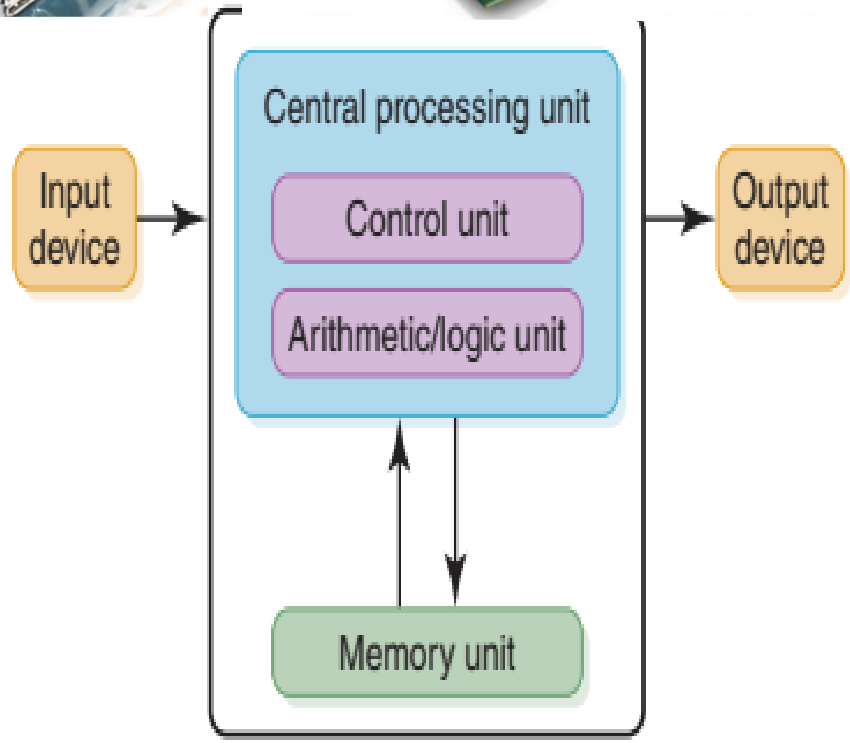
Contains the instruction that is being executed

Program counter (PC)

Contains the address of the next instruction to be executed

Central Processing Unit **(CPU)**

ALU and the control unit called the, or CPU

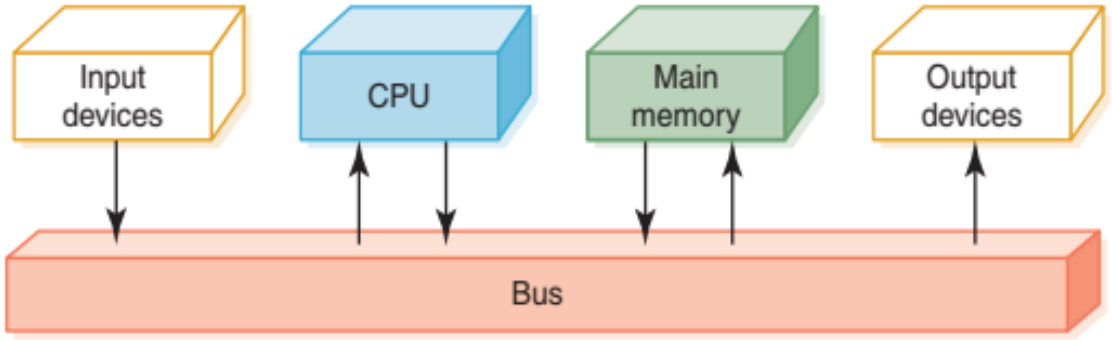


The von Neumann architecture

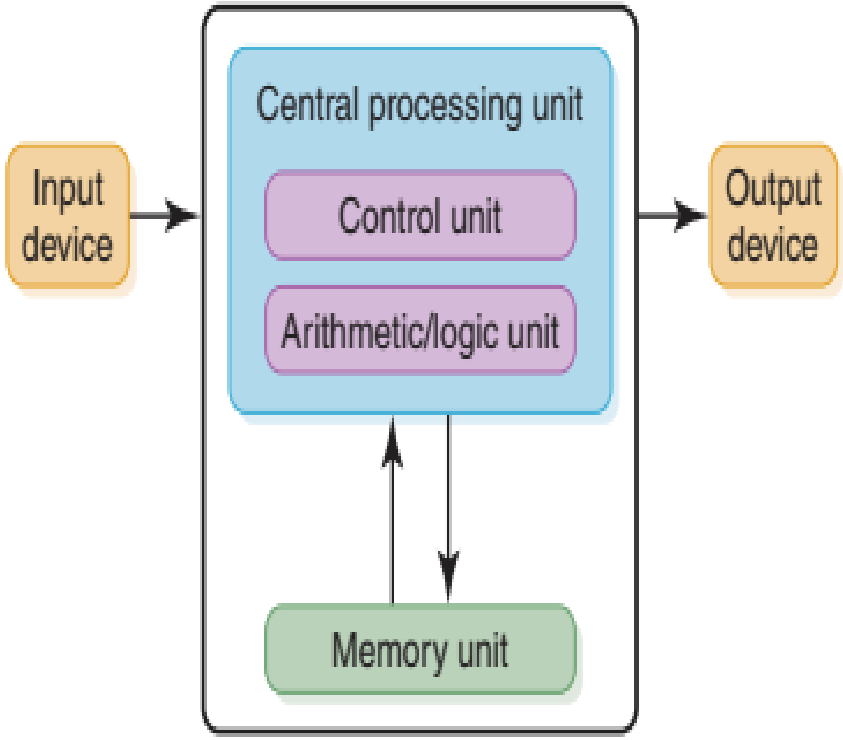
Flow of Information

Bus

A set of wires that connect all major sections

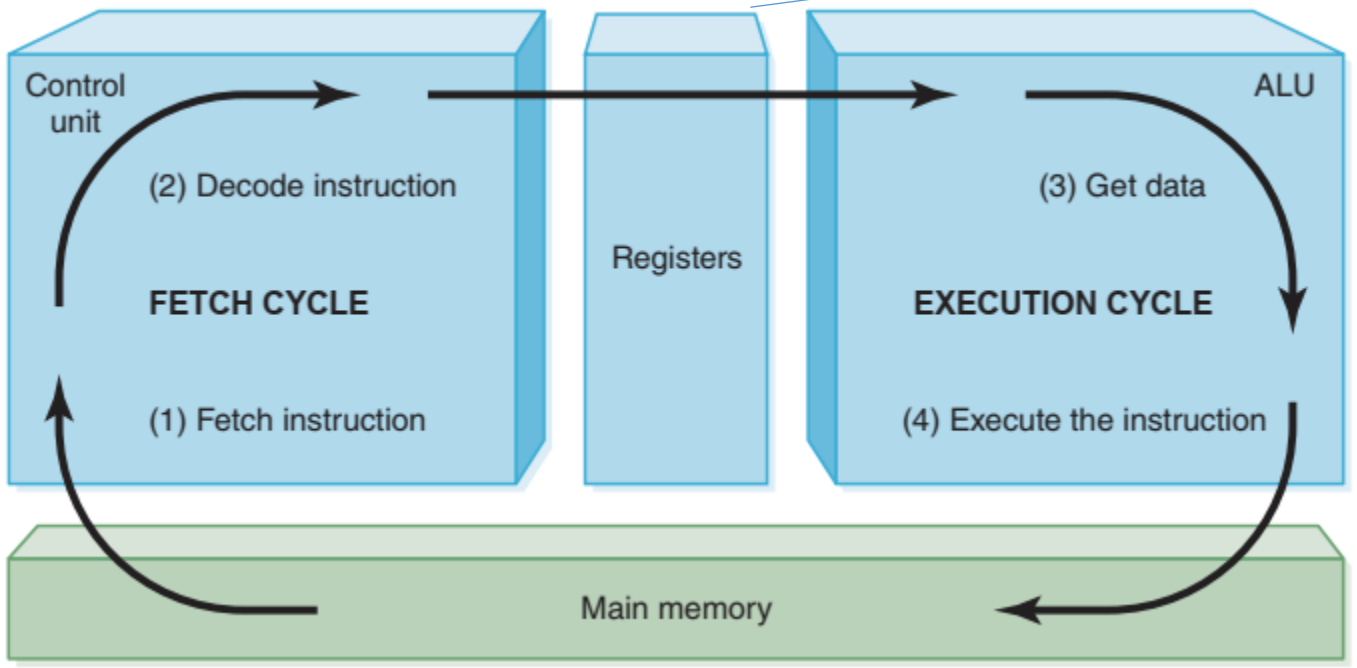


Data flow through a Von Neumann architecture



The von Neumann architecture

The Fetch–Execute Cycle



Instruction register (IR) The register that contains the instruction currently being executed

Program counter (PC) The register that contains the address of the next instruction to be executed

Fetch the next instruction
Decode the instruction
Get data if needed
Execute the instruction

The Programming Layer

- What's the layer is about?

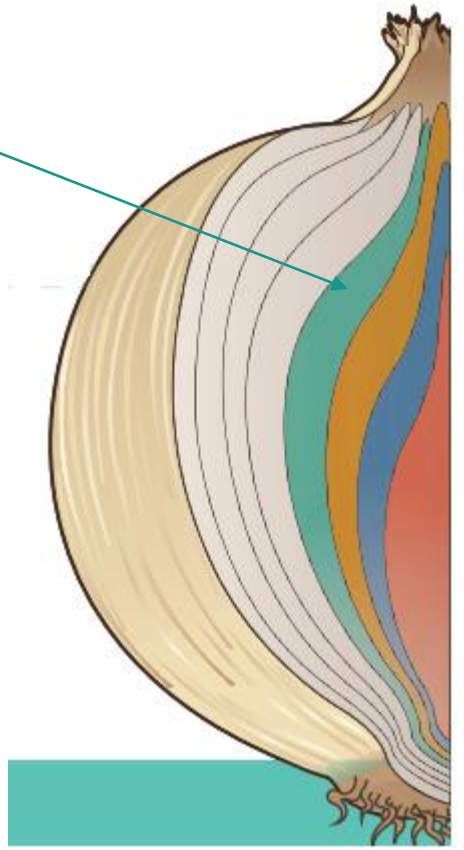
Programming Layer

How to use
a computer system

Hardware Layer

Information Layer

What a
computer system is

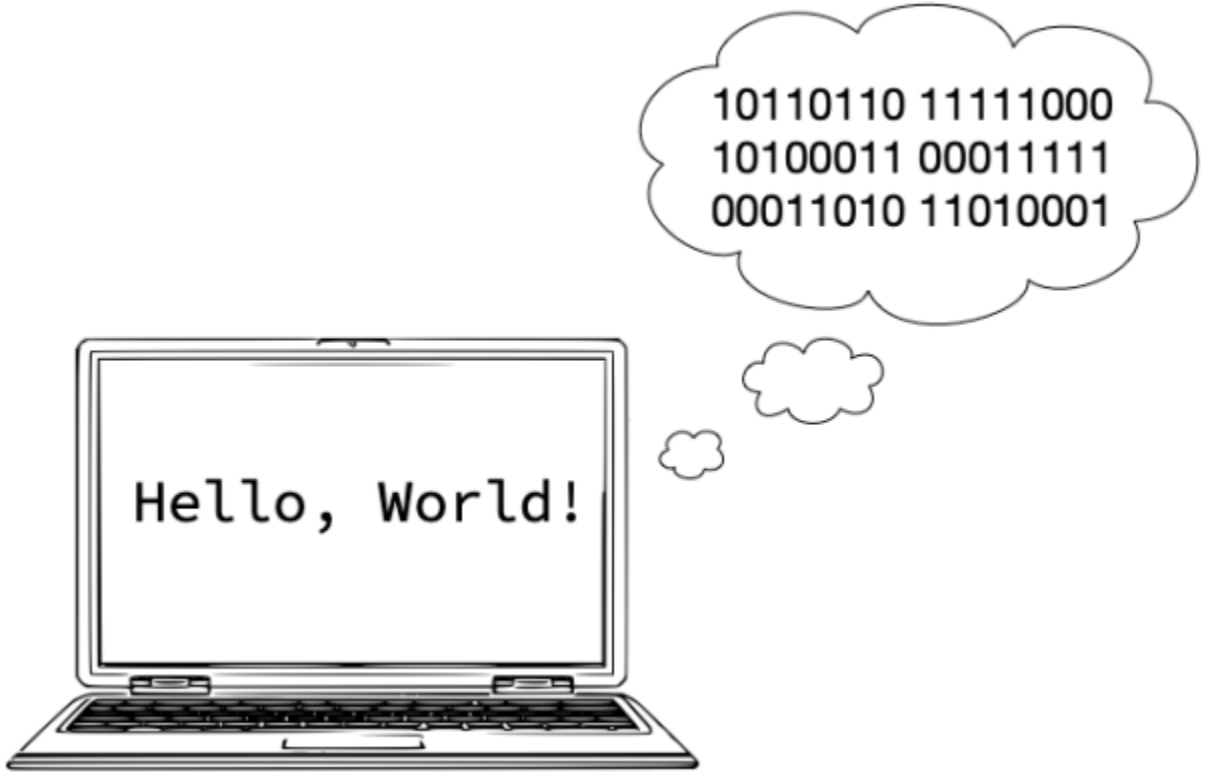


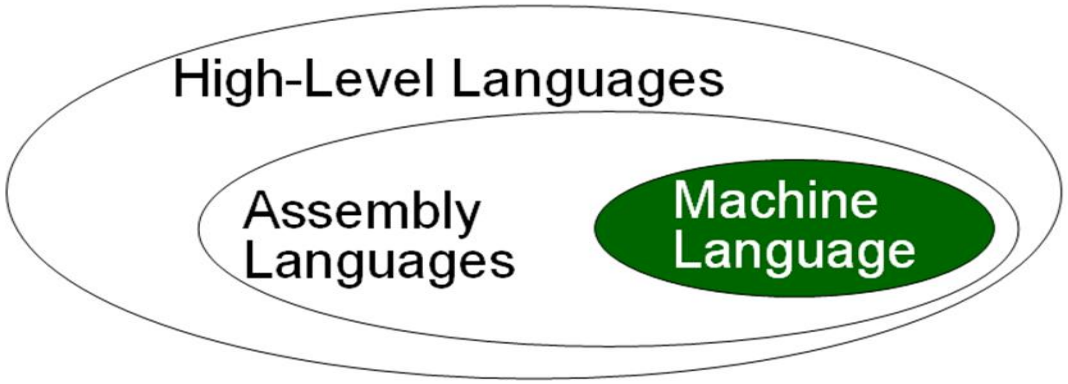
Programming

program: A set of instructions to be carried out by a computer.

program execution: The act of carrying out the instructions contained in a program.

programming language: A systematic set of rules used to describe computations in a format that is editable by humans.





machine code

```
10110100 10110111 00101011
00011010 00010100 10111011
10001000 11110111 00101000
10101010 00101101 00010001
01010010 11101100 11010001
10010100 10010100 00100000
00000000 10100001 00110001
10101001 00010101 00101010
00100100 10010100 01110001
11110101 11101011 00101111
01010010 10000101 11111110
10101010 00101101 00010001
01010010 11101100 11010001
10010100 10010100 00100000
00000000 10100001 00110001
10101001 00010101 00101010
00100100 10010100 01110001
11110101 11101011 00101111
01010010 10000101 11111110
00101001 00000000 00000000
00000000 00000000 01010000
00001010 00001010 00101010
00101010 00100100 10011111
```

assembly

```
.equ STDOUT, 1
.equ SVC_WRITE, 64
.equ SVC_EXIT, 93

.text
.global _start
_start:
    stp x29, x30, [sp, -16]!
    mov x0, #STDOUT
    ldr x1, =msg
    mov x2, 13
    mov x8, #SVC_WRITE
    mov x29, sp
    svc #0
    ldp x29, x30, [sp], 16
    mov x0, #0
    mov x8, #SVC_EXIT
    svc #0

msg:  .ascii "Hello World!\n"
.align 4
```

C++

```
#include <iostream>

int main () {
  std::cout << "Hello World!"
     << std::endl;
}
```

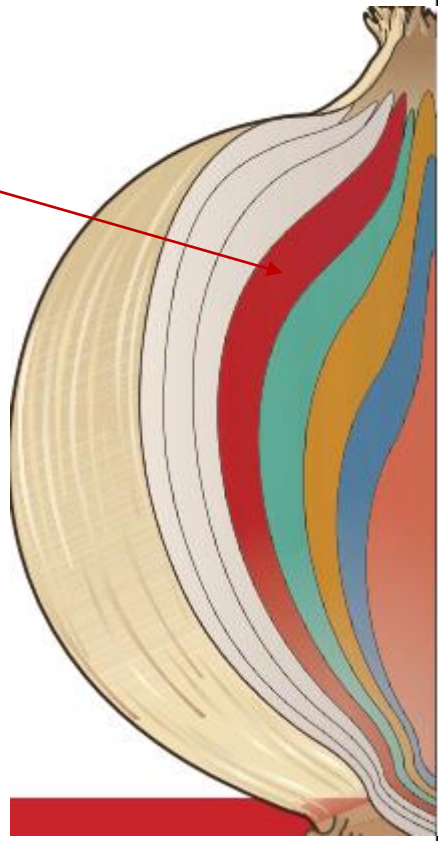
Python

```
print('Hello World')
```

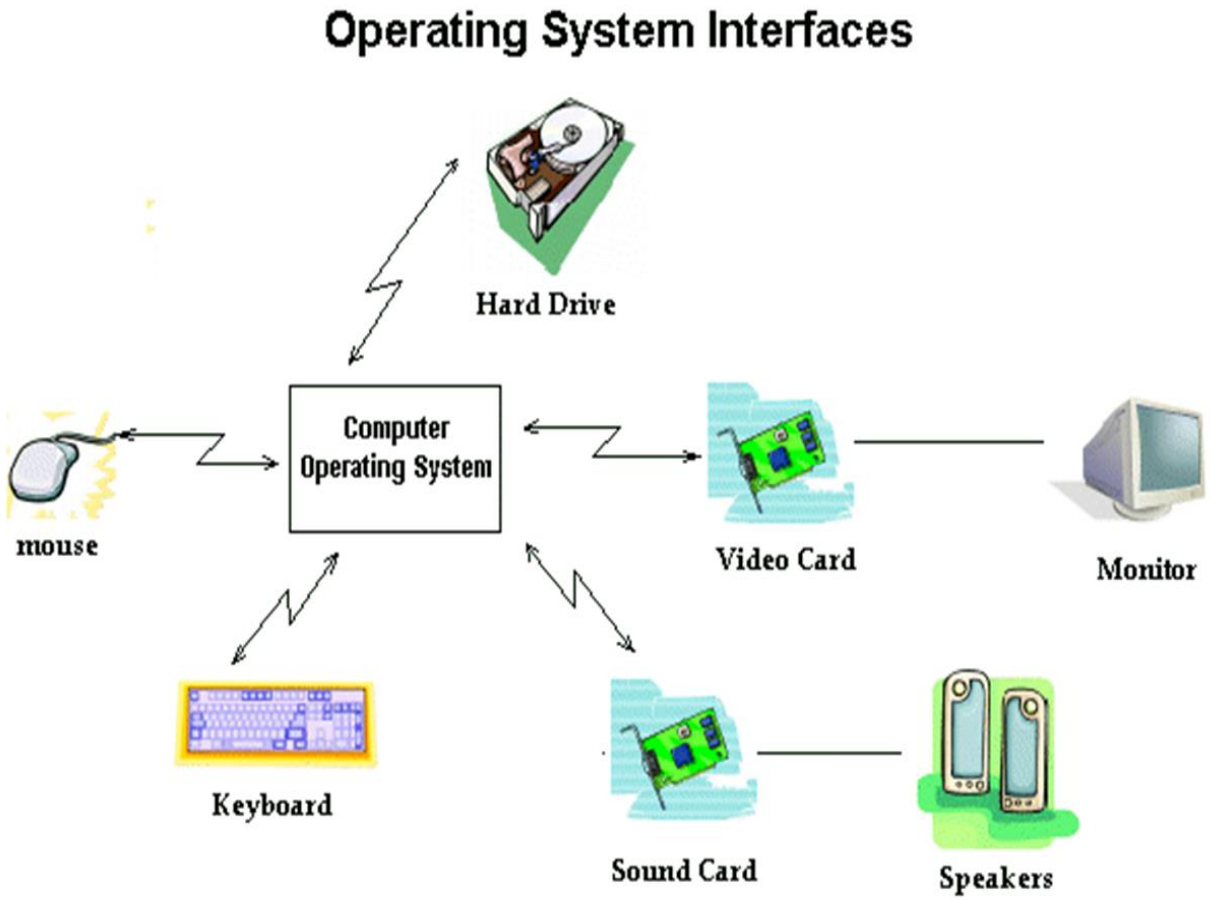
increasing abstraction →

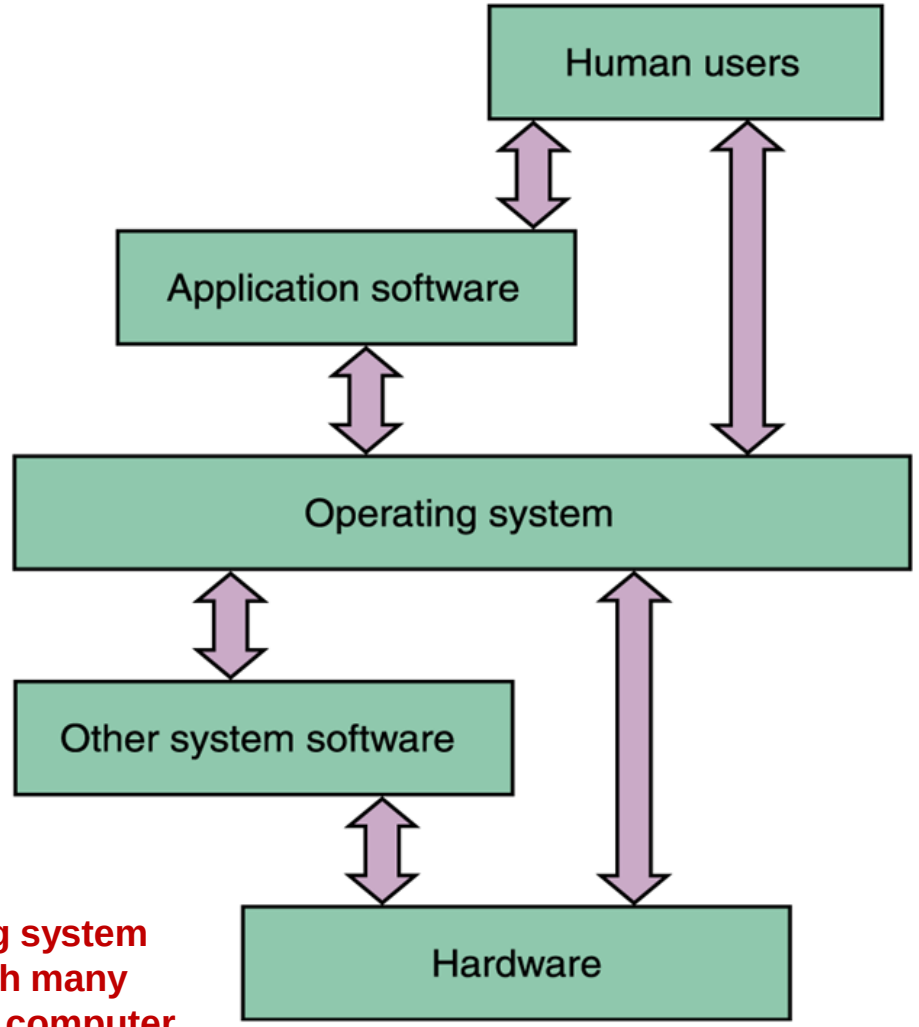
The Operating Systems Layer

- Operating System (OS) is the core of system software that
  - manages resources: memory, I/O devices.
  - provides an interface between users and computers.
  - allows an application to interact with system resources



Software Category





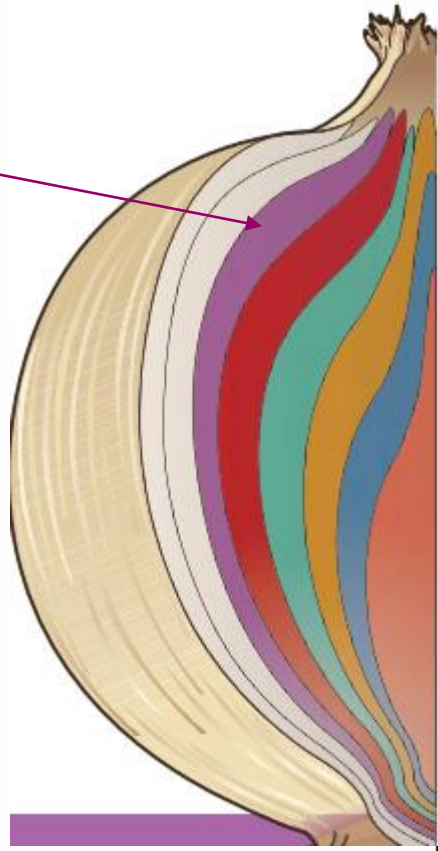
An operating system interacts with many aspects of a computer system.

UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

The Applications Layer

Application Software

- Consists of programs designed to make users more productive and/or assist with personal tasks.
  - To make business activities more efficient
    - Microsoft Office: Word, Excel, PowerPoint, Access, One Note, InfoPath, Outlook,…
  - To assist with graphics and multimedia projects
    - AutoCAD, Adobe Photoshop, …
  - To support home, personal, and educational tasks
    - Google map, Google Classroom, …
  - To facilitate communications
    - Web browser (IE, Firefox Chrome, Safari, Opera,…), Gmail, Zalo, Messenger, Skype …

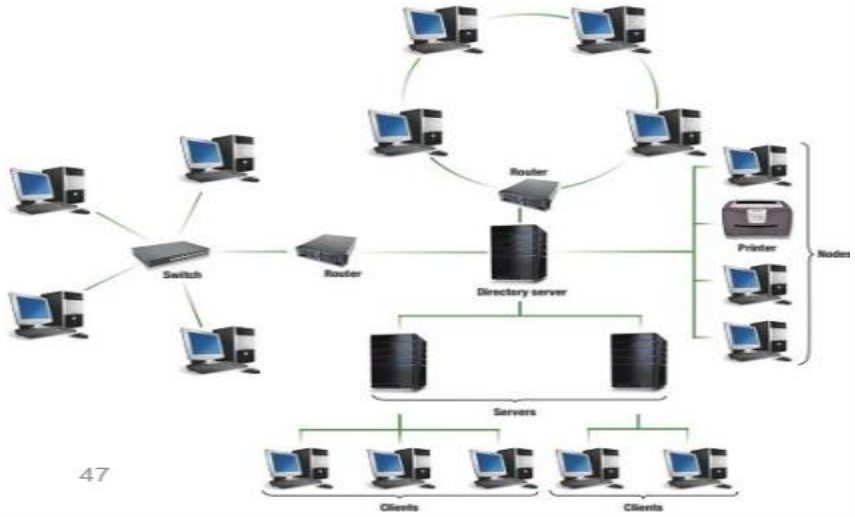


UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

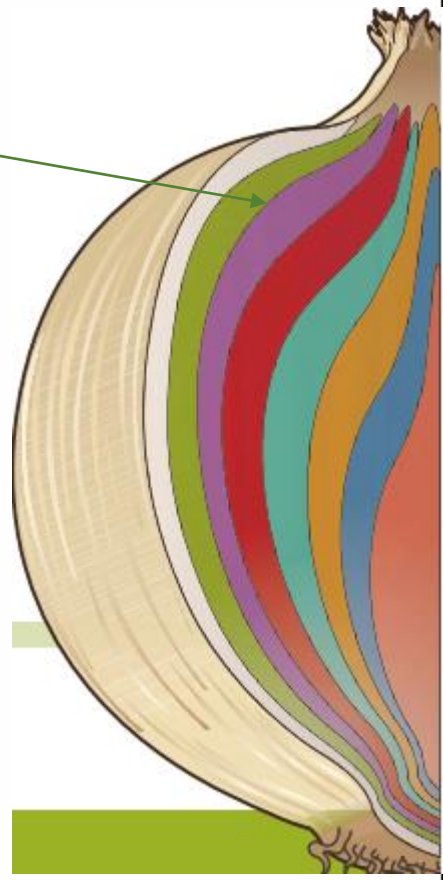
The Communications Layer Networks

- **A computer network** is a collection of computers and devices connected together via communications devices and transmission media.
- Networks allow computers to share resources such as hardware, software, data, and information.





47



Q&A